Geometry of Data - HW 2

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Question-1 Geodesic equation on S^2

(a) $$S(\theta,\phi) = (\cos\theta\cos\phi, \sin\theta\cos\phi, \sin\phi)$$

$$\frac{\partial S}{\partial \theta} = (-\sin\theta\cos\phi, \cos\theta\cos\phi, 0) \longrightarrow v^1$$

$$\frac{\partial S}{\partial \phi} = (-\cos\theta\sin\phi, -\sin\theta\sin\phi, \cos\phi) \longrightarrow v^2$$

(b) Riemannian metric $\longrightarrow$ metric tensor

$$g_{ij} = \langle v^i, v^j \rangle$$

$$\begin{aligned} g_{11} = \langle v^1, v^2 \rangle &= (-\sin\theta\cos\phi, \cos\theta\cos\phi, 0)\cdot(-\sin\theta\cos\phi, \cos\theta\cos\phi, 0) \\ &= \sin^2\theta\cos^2\phi + \cos^2\theta\cos^2\phi + 0 \\ &= \cos^2\phi\,\underbrace{(\sin^2\theta + \cos^2\theta)}_{1} \\ &= \cos^2\phi \end{aligned}$$

$$\begin{aligned} g_{12} = g_{21} = \langle v^1, v^2 \rangle &= (-\sin\theta\cos\phi, \cos\theta\cos\phi, 0)\cdot(-\cos\theta\sin\phi, -\sin\theta\sin\phi, \cos\phi) \\ &= \sin\theta\cos\theta\cos\phi\sin\phi - \sin\theta\cos\theta\cos\phi\sin\phi \\ &= 0 \end{aligned}$$

$$\begin{aligned} g_{22} = \langle v^2, v^2 \rangle &= (-\cos\theta\sin\phi, -\sin\theta\sin\phi, \cos\phi)\cdot(-\cos\theta\sin\phi, -\sin\theta\sin\phi, \cos\phi) \\ &= \cos^2\theta\sin^2\phi + \sin^2\theta\sin^2\phi + \cos^2\phi \\ &= \sin^2\phi(\cos^2\theta + \sin^2\theta) + \cos^2\phi \\ &= \sin^2\phi + \cos^2\phi = 1 \end{aligned}$$

$$g = \begin{bmatrix} \cos^2\phi & 0 \\ 0 & 1 \end{bmatrix} \xrightarrow{\quad} g_{ij}$$

$\rightarrow$ metric tensor of the riemannian metric

(c)

$$\text{if } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \qquad A^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & -a \end{bmatrix}$$

$$g^{-1} = \frac{1}{\cos^2\phi - 0} \cdot \begin{bmatrix} 1 & 0 \\ 0 & \cos^2\phi \end{bmatrix} = \begin{bmatrix} \sec^2\phi & 0 \\ 0 & 1 \end{bmatrix}$$

$$g^{-1} = \begin{bmatrix} \sec^2\phi & 0 \\ 0 & 1 \end{bmatrix} \longrightarrow g^{ij}$$

(d) Christoffel symbols $\Gamma^k_{ij} = \frac{1}{2}\sum_{l=1}^{n} g^{kl}\left(\frac{\partial g_{jl}}{\partial x^i} + \frac{\partial g_{il}}{\partial x^j} - \frac{\partial g_{ij}}{\partial x^l}\right)$ $\quad x^1 = \theta$, $x^2 = \phi$

$$\Gamma^1_{12} = \frac{1}{2}\left(g^{11}\left(\frac{\partial g_{11}}{\partial x^1} + \frac{\partial g_{11}}{\partial x^1} - \frac{\partial g_{11}}{\partial x^1}\right) + \underset{\downarrow\, 0}{g^{12}}\left(\cdot\right)\right)$$

$\rightarrow$ inconsequential

$$= \frac{1}{2}\left(\sec^2\phi\left(\frac{\partial(\cos^2\phi)}{\partial\theta} + \frac{\partial(\cos^2\phi)}{\partial\theta} - \frac{\partial(\cos^2\phi)}{\partial\theta}\right)\right)$$

$$= 0$$

$$\Gamma^1_{12} = \frac{1}{2}\left(g^{11}\left(\frac{\partial g_{21}}{\partial x^1} + \frac{\partial g_{11}}{\partial x^2} - \frac{\partial g_{12}}{\partial x^1}\right) + \underset{\downarrow\, 0}{g^{12}}\left(\cdot\right)\right)$$

$$= \frac{1}{2}\left(\sec^2\phi\left(0 + \frac{\partial(\cos^2\phi)}{\partial\phi} - 0\right)\right.$$

$$= \frac{1}{2}\sec^2\phi \cdot 2\cos\phi(-\sin\phi)$$

$$= -\sec\phi\sin\phi = -\tan\phi$$

$$\Gamma_{21}^{\;1} = \frac{1}{2}\left(g^{11}\left(\frac{\partial g_{11}}{\partial x^2} + \frac{\partial g_{21}}{\partial x^1} - \frac{\partial g_{21}}{\partial x^1}\right) + g^{12}\overset{\to 0}{(\cdot)}\right)$$

$$= \frac{1}{2}\left(\sec^2\phi\left(\frac{\partial \cos^2\phi}{\partial \phi} + 0 - 0\right)\right)$$

$$= \frac{1}{2}\left(\sec^2\phi\,(2\cos\phi)(-\sin\phi)\right)$$

$$= -\tan\phi$$

$$\Gamma_{22}^{\;1} = \frac{1}{2}\left(g^{11}\left(\frac{\partial g_{21}}{\partial x^2} + \frac{\partial g_{21}}{\partial x^2} - \frac{\partial g_{22}}{\partial x^1}\right) + g^{12}\overset{\to 0}{(\cdot)}\right)$$

$$= \frac{1}{2}\left(\sec^2\phi\,(0+0-0)\right)$$

$$= 0$$

from the above calculated christoffel symbols we can observe that only the partial derivative of $\frac{\partial g_{11}}{\partial x^2}$ is non-zero, all others are zero

$$\Gamma_{11}^{\;2} = \frac{1}{2}\left(g^{21}\overset{\to 0}{(\cdot)} + g^{22}\left(0+0-\frac{\partial g_{11}}{\partial x^2}\right)\right)$$

$$= \frac{1}{2}\left(1\cdot\left(-\frac{\partial \cos^2\phi}{\partial \phi}\right)\right)$$

$$= \frac{1}{2}\cdot 1 \cdot -1 \cdot 2\cos\phi\,(-\sin\phi)$$

$$= \sin\phi\cos\phi$$

$$\Gamma_{12}^{\;2} = \Gamma_{21}^{\;2} = \Gamma_{22}^{\;2} = 0$$

we can deduce this through observation since all of the terms end up to zero because g^{21} is 0 and g^{22} means that $l = 2$ and the only possibility of $\frac{\partial g_{11}}{\partial x^2}$ occurring is in $\Gamma_{\underset{i}{1}\underset{j}{1}}^{\;2}$ when $l = 2$

therefore, now the non-zero christoffel symbols are

$$\Gamma_{12}^{1} = \Gamma_{21}^{1} = -\tan\phi \ , \ \Gamma_{11}^{2} = \sin\phi\cos\phi$$

(e)

We have an equator circle $\gamma(t) = S(t,0)$

for a curve to be a geodesic, it has to satisfy the differential equation

$$\frac{d^2\gamma^k}{dt^2} = -\sum_{i,j=1}^{n} \Gamma_{ij}^{k}(\gamma(t))\frac{d\gamma^i}{dt}\frac{d\gamma^j}{dt} \quad \text{where } i,j = \{1,2\}$$

the only non-zero $\frac{d\gamma^i}{dt}$ is $\frac{d\gamma^1}{dt} = \frac{dt}{dt} = 1$, however

$$\frac{d^2\gamma^1}{dt^2} = \frac{d^2\gamma^2}{dt^2} = 0$$

$$\downarrow \qquad \downarrow$$

$$\frac{d^2(t)}{dt^2} \qquad \frac{d^2(0)}{dt^2}$$

for any value of k at $\phi = 0$ the non-zero Christoffel symbols $\Gamma_{12}^{1} = \Gamma_{12}^{1} = -\tan\phi$, $\Gamma_{11}^{2} = \sin\phi\cos\phi$. these christoffel symbols become zero since $\tan 0 = 0$ and $\sin 0 = 0$

therefore since all of the terms of the above differential equation are zero for the equator $S(t,0)$, the equation is satisfied. Therefore the given equator is a geodesic

(f) a great circle is a circle on the surface of the sphere that has radius equal to radius of the sphere, this should mean that the plane of the circle is always passing through the origin just like an equator. if you think of it, any great circle on the sphere can be made into an equator of the sphere by rotating the axes accordingly and since we proved that the equator is a geodesic in the previous part of the question this implies that any great circle on the sphere is also a geodesic.